DSA0503 – Query Processing for Data Science

CO2_AT2_Schema Analysis

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Experiment 1: Library Management Database

Aim

To use Python basics and the SQLite (sqlite3) library to connect to a database and retrieve the member name, book title, and issue date for all books issued after **1 January 2026**.

Requirement Analysis

A library maintains the following database schema:

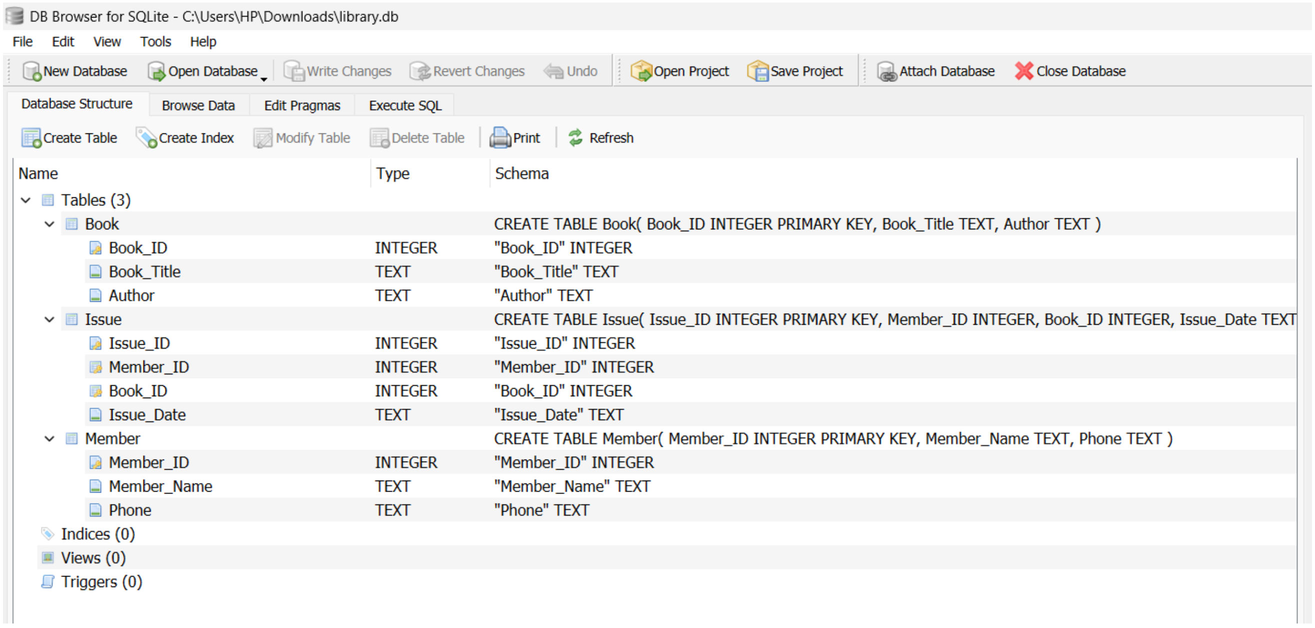
- **Member (Member_ID, Member_Name, Phone)**
- **Book (Book_ID, Book_Title, Author)**
- **Issue (Issue_ID, Member_ID, Book_ID, Issue_Date)**

The objective is to retrieve:

- Member Name
- Book Title
- Issue Date

for all books issued after **01-01-2026**.

Database Schema



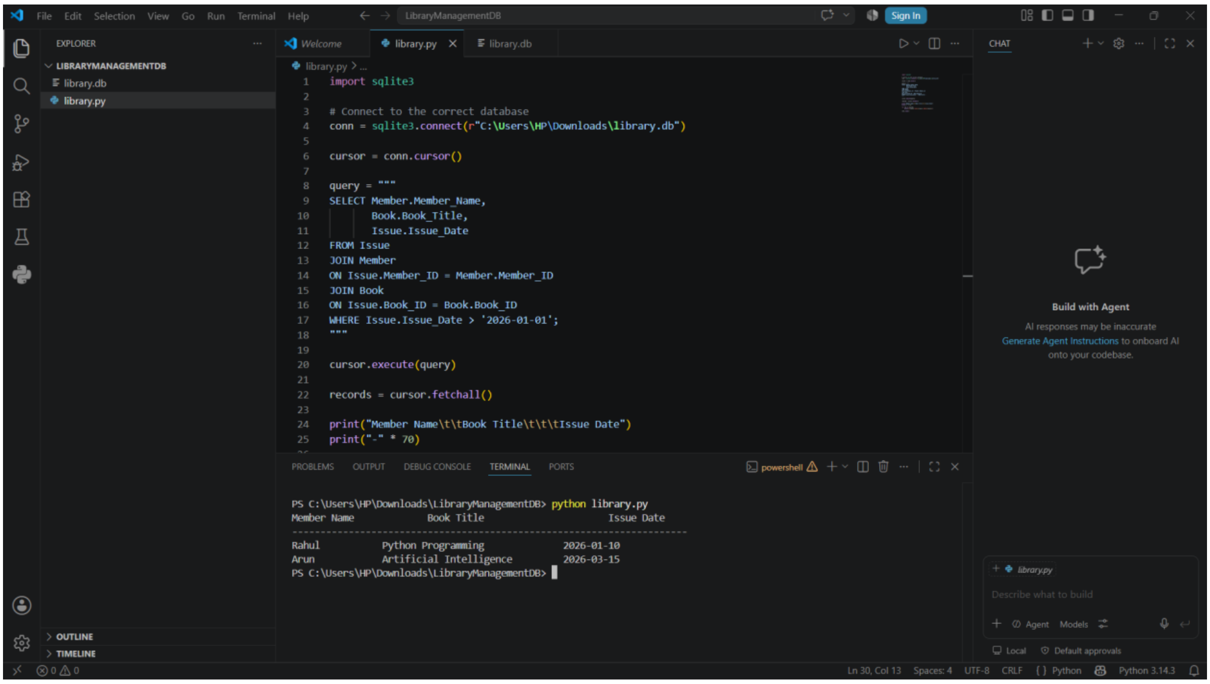
SQL Query

```
SELECT Member.Member_Name,
       Book.Book_Title,
       Issue.Issue_Date
FROM Issue
JOIN Member
ON Issue.Member_ID = Member.Member_ID
JOIN Book
ON Issue.Book_ID = Book.Book_ID
WHERE Issue.Issue_Date > '2026-01-01';
```

Python Program

```python
import sqlite3

# Connect to the correct database
conn = sqlite3.connect(r"C:\Users\HP\Downloads\library.db")

cursor = conn.cursor()

query = """
SELECT Member.Member_Name,
       Book.Book_Title,
       Issue.Issue_Date
FROM Issue
JOIN Member
ON Issue.Member_ID = Member.Member_ID
JOIN Book
ON Issue.Book_ID = Book.Book_ID
WHERE Issue.Issue_Date > '2026-01-01';
"""

cursor.execute(query)

records = cursor.fetchall()

print("Member Name\t\tBook Title\t\t\tIssue Date")
print("-" * 70)

for row in records:
    print(f"{row[0]:<15}\t{row[1]:<30}\t{row[2]}")

conn.close()
```

Output



Result

The SQLite database was successfully connected using Python (sqlite3). The required records were retrieved by joining the **Member**, **Book**, and **Issue** tables. The program displayed the member name, book title, and issue date for books issued after **1 January 2026**.

Conclusion

The experiment was completed successfully using Python and SQLite. Database connectivity was established, SQL JOIN operations were performed, and the required records were retrieved based on the specified date condition.

Experiment 2: Hotel Reservation Database

Aim

To use Python basics and the SQLite (`sqlite3`) library to connect to a database and retrieve the customer name, room type, and check-in date for customers who have reserved **Deluxe** rooms.

Requirement Analysis

A hotel maintains the following database schema:

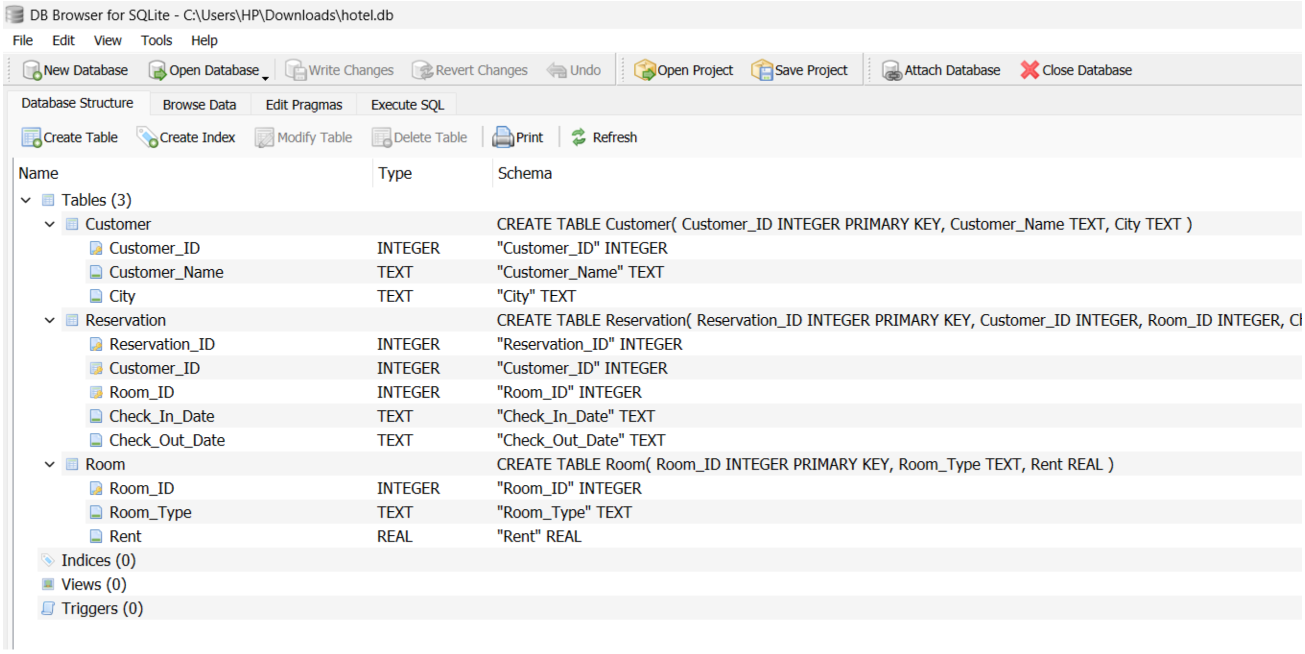
- **Customer (Customer_ID, Customer_Name, City)**
- **Room (Room_ID, Room_Type, Rent)**
- **Reservation (Reservation_ID, Customer_ID, Room_ID, Check_In_Date, Check_Out_Date)**

The objective is to retrieve:

- Customer Name
- Room Type
- Check-In Date

for customers who have reserved **Deluxe** rooms.

Database Schema



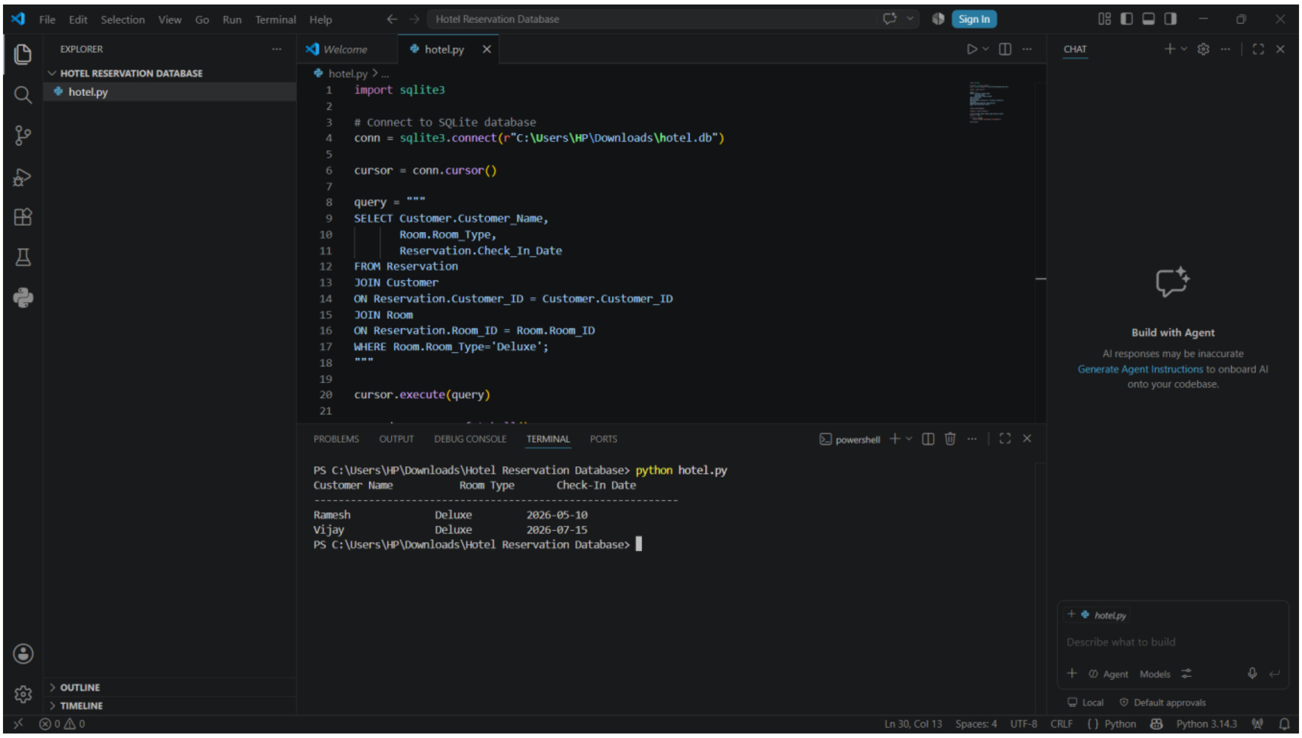
SQL Query

```
SELECT Customer.Customer_Name,
       Room.Room_Type,
       Reservation.Check_In_Date
FROM Reservation
JOIN Customer
ON Reservation.Customer_ID = Customer.Customer_ID
JOIN Room
ON Reservation.Room_ID = Room.Room_ID
WHERE Room.Room_Type = 'Deluxe';
```

Python Program

```python
import sqlite3

# Connect to SQLite database
conn = sqlite3.connect(r"C:\Users\HP\Downloads\hotel.db")

cursor = conn.cursor()

query = """
SELECT Customer.Customer_Name,
       Room.Room_Type,
       Reservation.Check_In_Date
FROM Reservation
JOIN Customer
ON Reservation.Customer_ID = Customer.Customer_ID
JOIN Room
ON Reservation.Room_ID = Room.Room_ID
WHERE Room.Room_Type='Deluxe';
"""

cursor.execute(query)

records = cursor.fetchall()

print("Customer Name\t\tRoom Type\tCheck-In Date")
print("-" * 60)

for row in records:
    print(f"{row[0]:<20}{row[1]:<15}{row[2]}")

conn.close()
```

Output



Result

The SQLite database was successfully connected using Python (`sqlite3`). The required records were retrieved by joining the **Customer**, **Room**, and **Reservation** tables. The program displayed the customer name, room type, and check-in date for customers who reserved **Deluxe** rooms.

Conclusion

The experiment was completed successfully using Python and SQLite. Database connectivity was established, SQL `JOIN` operations were performed, and the required records were retrieved based on the room type **Deluxe**.